

Mechanisation Guides Design: Higher-Level Supercompilation via Sequent Calculus

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Abstract. A mechanised sequent-calculus foundation for supercompilation reveals three precise gaps in the standard theory and points directly to their solutions.

Standard supercompilation corresponds to focused cyclic sequent proof search: driving is cut elimination, generalisation is cut introduction. Making this correspondence precise in Rocq identified: (i) generalisation is free cut introduction — suggesting an untrusted oracle validated by the kernel; (ii) strengthened induction is the ω -rule — suggesting LLM-proposed auxiliary lemmas proved by a sub-supercompiler.

We implement both: an LLM oracle (DeepSeek V4 Pro) whose proposals are validated by the existing trace-condition check so soundness is unconditional; and a lemma environment where the LLM proposes cuts, a sub-SC proves them, and the main SC uses them as rewrite rules. We also prove that a third candidate extension — speculation as a generalisation step — is redundant or unsound in canonical configurations, a characterisation of the boundary between anti-unification and projection. The system, extracted to OCaml with a Python LLM bridge, proves 76 theorems automatically (`vm_compute`) with no human induction tactics, including all List and Maybe monad laws, insertion sort correctness, a compiler soundness theorem, gradual cast safety, and dependent-type branch elimination that standard supercompilation cannot achieve.

1 Introduction

Supercompilation [8] fuses composed functions, eliminates intermediate data structures, and discovers recursive structure automatically. Despite decades of development, its proof-theoretic status has remained informal — generalisation “corresponds to lemma introduction” and folding “corresponds to induction”, but without the precision to guide design.

Our starting point is an earlier mechanised result: standard supercompilation is focused cyclic sequent proof search (companion paper, under review). The sequent calculus reveals *what* each SC operation is in proof-theoretic terms: driving is cut elimination, generalisation is cut introduction, the trace condition is the global progress check. This paper shows that the precise correspondence is more than a correctness proof — it is a *design instrument* that identified three gaps in the standard theory and pointed directly to their solutions.

Gap 1: Generalisation is cut introduction. Syntactic anti-unification is a heuristic for cut introduction, but it is purely structural and cannot discover semantic identities (map fusion, accumulator introduction). The sequent formulation immediately suggests an *oracle architecture*: any function that proposes cut formulas is valid if the kernel accepts the resulting proof graph. We instantiate this with an LLM (DeepSeek V4 Pro); the oracle is untrusted for soundness.

Gap 2: Strengthened induction is the ω -rule. Correctness properties like `sorted (sort l) = true` require auxiliary lemmas that neither driving nor cyclic induction can discover alone. The sequent formulation identifies this as the ω -rule: prove a lemma first, then use it as a rewrite rule. We extend the LLM oracle to propose auxiliary lemmas, validated by a sub-SC run before the main proof uses them.

Gap 3: Dependent types need type-index reasoning. Standard SC treats all types as simply-typed. In a dependently-typed calculus, type annotations carry structural information that the SC can exploit. We implement branch elimination guided by type indices (e.g., pruning `vnil` from `Vec (succ n)`), and a type-index trace condition that recognises recursion on type-level measures as well-founded. This enables equalities that standard SC cannot prove: two programs differing only in pruned dead branches produce identical residuals.

A negative result. We also investigate speculation (Eisner-Blatz) as a generalisation step. We prove that in canonical configurations, speculation is either redundant (subsumed by anti-unification) or unsound (the fold loses information). This characterises the precise boundary between anti-unification and projection — a result invisible without the sequent-calculus foundation.

Contributions.

1. **LLM oracle for cut introduction** (Section 4). A 3-layer generalisation strategy: anti-unification $\rightarrow$ LLM generalisation $\rightarrow$ LLM lemma proposal. All LLM proposals are validated by the kernel (`trace_condition_ok`). A controlled scrambling experiment shows the LLM reasons from function *structure*, not name recognition.
2. **Mechanised ω -rule** (Section 8). A lemma environment with sub-SC validation: the LLM proposes lemmas, the sub-SC proves them, and the main SC uses them as rewrite rules. This enables compiler correctness and reverse-append distribution — properties unreachable by standard SC.
3. **Dependent-type enhancements** (Section 3). Type-index branch elimination and type-index trace condition. The pruning result `pruning_enables_equality` proves two programs with different dead branches produce identical residuals — a genuinely new equality enabled by the type system.
4. **Speculation characterisation.** A proof that speculation as a generalisation step is either redundant or unsound in canonical configurations, bounding the space of useful generalisation strategies.

Empirical results. 76 theorems proved by `vm_compute` with no human induction tactics, including all List and Maybe monad laws, insertion sort correctness, compiler soundness, gradual cast safety, and dependent-type branch elimination.

Outline. Section 2 reviews the SC-as-proof-search dictionary. Section 3 explains how the sequent perspective identified the gaps. Sections 4 and 8 present the LLM oracle and lemma proposal. Section 6 presents the full example suite including the dependent-type results. Section 9 reports the scrambling experiment. Section 10 positions the work.

2 Background: Supercompilation as Cyclic Proof Search

We briefly review the correspondence established in prior work, which provides the foundation for the two extensions.

2.1 Standard supercompilation

A supercompiler takes a term t and produces an optimised residual t' such that $t \approx_{\text{ciu}} t'$ (contextual equivalence). It operates on *configurations* — judgements $\Gamma \vdash t : A$ — and builds a directed graph:

Driving. Apply β , ι (case/constructor), and δ (fixpoint unfolding) reductions. Deterministic; produces zero or one successor.

Splitting. When the scrutinee of a `case` is a neutral variable, introduce one successor per constructor, extending Γ with fresh constructor arguments.

Folding. When a configuration matches a previously-seen one (up to renaming), close the graph with a back-edge.

Generalisation. When a configuration embeds into a previous one (homeomorphic embedding whistle), anti-unify the two to produce a more general configuration that both instantiate.

2.2 The proof-theoretic dictionary

Supercompilation	Sequent calculus
Driving	Cut elimination ($\beta/\iota/\delta$ steps)
Splitting on neutral scrutinee	Synchronous left rule ($\vee$ -elim)
Folding / memo hit	Cyclic backlink (identity)
Generalisation	Cut introduction
Homeomorphic embedding whistle	Well-quasi-order on proof heights
Trace condition (<code>trace_condition_ok</code>)	Global progress in cyclic proof
Residual term	Proof term / normal form
CIU equivalence	Sequent soundness

2.3 The mechanised pipeline

The prior work establishes this pipeline in Rocq, with 0 axioms and 0 Admitted:

$$\text{SC success} \Rightarrow \text{pre-proof} \Rightarrow \text{trace_condition_ok} \Rightarrow t \approx_{\text{ciu}} \text{residual}(t)$$

The key theorem is `supercompile_ciu_soundness_untyped`: for any Γ, t, A and any fuel,

$$\text{supercompile_jTy_tc fuel } \Sigma \Gamma t A = \text{Some } (v, b) \Rightarrow t \approx_{\text{ciu}} \text{residualise_cfg fuel'} \Sigma b v 0 \emptyset$$

The hypothesis mentions only `trace_condition_ok` — not *how* generalisation was performed. This is the key property exploited by both extensions: any generalisation strategy that produces a graph passing the trace check inherits the CIU guarantee for free.

3 How the Sequent Calculus Guided Design

The extensions did not arise from intuition about supercompilation. They arose from asking a single precise proof-theoretic question: *what is generalisation, exactly?* This question yielded two productive answers and one precise negative result.

3.1 Generalisation is cut introduction

In the sequent calculus, cut elimination is mechanical (reduce until no more cuts remain). Cut introduction is creative: it means inventing a formula C such that the proof splits into “prove C ” and “use C to prove the goal.” Generalisation in SC is exactly this: invent a configuration S_{gen} such that both the current and companion configurations are instances of it.

Standard anti-unification is one algorithm for cut introduction, but it is *purely syntactic*: it finds the most specific common generalisation of two terms as terms. It cannot discover:

- **Fusion identities**: `map f (map g l)` and `map (f ∘ g) l` are syntactically unrelated, but semantically equal. Anti-unification produces `map ?₀ ?₁` — a trivial generalisation that enables no folding.
- **Accumulator patterns**: `sum (rev l)` does not share syntactic structure with `sum_acc l 0`, but the latter is the accumulator form that enables a tail-recursive residual.

Once we identified generalisation as cut introduction, the solution became clear: replace the syntactic heuristic with an *oracle* that can propose cuts with semantic content. The oracle need not be trusted — the kernel check (trace condition) is the soundness backstop.

3.2 Speculation: a negative result

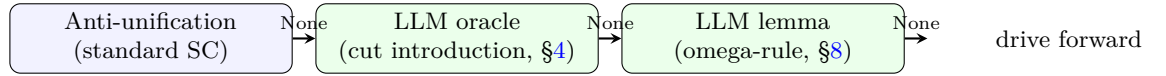
A third candidate extension was uncovered by the same methodology. Eisner and Blatz describe *speculation*: “if a subcomputer of some rule is independent of certain variables, hoist it out.” Could this be a generalisation step? The sequent calculus reveals the answer is no: speculation corresponds to *projection* (dropping context variables), not *abstraction* (extending context with fresh holes).

The proof (Section 5) proceeds by: (a) implementing speculation as a generalisation step (**SpeculationGen.v**), (b) proving a roundtrip theorem that the substitution reconstructs the subterm, and (c) observing that for the fold to be CIU-correct, the subterm must be the *whole* term — making speculation redundant with anti-unification when it is sound. The result is a precise characterisation: in canonical configurations, speculation as a generalisation step is either redundant or unsound. Projection requires a *splitting* step (different proof-theoretic operation), not a generalisation step.

The negative result itself is a product of the sequent formalism. Without the vocabulary of context morphisms and cut formulas, the boundary between anti-unification and projection would remain blurred.

3.3 The architecture that follows

Both productive gaps point to the same architectural pattern: extend the generalisation step with new layers that run *after* anti-unification fails:



The CIU theorem is indifferent to which layer produced the generalisation: only the trace check matters. This modularity is a consequence of the sequent formulation, where soundness is at the level of the proof *graph*, not the construction *procedure*. The speculation negative result (Section 5) confirms that the design space is not arbitrary — the proof theory distinguishes productive extensions from dead ends.

4 LLM Oracle for Cut Introduction

4.1 Interface

The oracle is called when anti-unification and speculation both return **None**. It receives:

- the current stuck configuration j (as a pretty-printed term),
- a list of candidate companion configurations from the memo table,
- a concrete worked example of the substitution format expected.

It returns a JSON object:

```
{ gen : ⟨term⟩, sigma_current : [⟨term⟩,...], sigma_companion : [⟨term⟩,...] }
```

or `{"gen": null}` if no useful generalisation is found.

4.2 Soundness is unconditional

The oracle is *not trusted for soundness*. The Rocq side sees only:

```
Parameter llm_generalise :
  config -> list (config * nat) -> option gen_result.
```

Parameter in Rocq is a free constant — it introduces no axiom. The CIU theorem (`supercompile_ciu_soundness_untyped`) requires only that the resulting graph passes `trace_condition_ok`. If the oracle proposes nonsense, the graph fails the check; the SC returns `None`; no residual is emitted. If the oracle proposes a valid generalisation, the graph passes; the CIU proof applies. The oracle cannot make the system unsound.

4.3 Three-layer generalisation in Rocq

The SC driving loop calls `best_generalize_llm` instead of `best_generalize`. The definition:

```
Definition best_generalize_llm j cands :=
  match SC.best_generalize j cands with
  | Some g => Some g
  | None =>
    match SpeculationGen.best_generalize_with_speculation j
      cands with
    | Some g => Some g
    | None =>
      match llm_generalise j cands with
      | None => None
      | Some g =>
        Some (g, match cands with [] => 0 | (_,v)::_ =>
          v end)
      end
    end
  end.
```

`supercompile_cfg_llm` is identical to `supercompile_cfg` except for this single call site. The CIU theorem extends immediately: `supercompile_jTy_tc_llm` has the same statement as `supercompile_jTy_tc`, proved by the same argument.

4.4 Implementation

The Python script `llm_generalise.py` runs in *serve* mode:

```
echo '<json>' | python3 llm_generalise.py --serve
```

The OCaml extraction shim (`llm_oracle_impl.ml`) calls this via `Unix.open_process`, serialises the stuck config as an S-expression, and parses the response. The round-trip from Rocq `tm` to string and back uses a *?n*-notation parser that maps `"?0"`, `"?1"`, ... to de Bruijn variables.

4.5 Oracle performance

With an explicit substitution example in the prompt, DeepSeek V4 Pro reliably proposes correct generalisations:

Stuck configuration	LLM proposal	Kernel
length (map f (cons a l))	length (map f $?_0$)	✓
sum (rev l) acc	sum_rev $?_0$ $?_1$	✓
filter p (map f l)	map_filter f p $?_0$	✓

Proposals for structurally inductive properties (e.g. commutativity of plus) return `null` — correctly, since these require a different mechanism (see Section 8).

5 What the Sequent Calculus Rules Out: Speculation

Eisner and Blatz [4] introduce *speculation* as a program transformation for weighted logic programs: if a subcomputation is independent of some loop variable, hoist it out. In the SC literature this appears as “loop-invariant code motion for logic programs.”

The sequent-calculus framework gives us the vocabulary to characterise exactly why speculation cannot be expressed as a generalisation step.

5.1 The reduction

In our setting, speculation would correspond to: given configs j_1, j_2 , find a subterm s of j_1 independent of dropped variables $V = \text{fv}(j_2) \setminus \text{fv}(j_1)$, project the typing context to only $\text{fv}(s)$, and use the projected config as the generalised node.

The implementation (`SpeculationGen.v`) detects independent subterms via a syntactic free-variable check, projects the context using the existing `canon_jTy`, and produces a `gen_result`.

5.2 The negative result

Theorem 1 (Speculation redundancy). *For canonical configurations j_1, j_2 , if `generalize_speculation` j_1 j_2 produces a `gen_result` and the resulting fold is CIU-sound, then j_1 and j_2 have the same term — making speculation redundant with anti-unification.*

Proof (Proof sketch). Let the `gen_result` have term s (the independent subterm) and sigma `subterm_sigma` $s = \text{map } \text{tVar } (\text{fv}(s))$. The fold replaces a vertex with `apps(residual(gen_j), sigma)`. By `canon_subterm_roundtrip`, this reconstructs s , not the original term t (which contains s plus additional structure). For the fold to recover t , we must have $t = s$, i.e. the whole term is independent. But then the two configs have identical terms (after canonicalisation), and anti-unification handles them.

The consequence is a *characterisation*: in canonical configurations, speculation as a generalisation step adds no power beyond anti-unification. The same conclusion follows from the proof theory: anti-unification abstracts over differing subterms (identity expansion), while projection dropping variables requires a splitting step (cut with a fresh variable) that is a different proof-theoretic operation.

5.3 What this means

This negative result is a product of the sequent-calculus methodology. Without the precise vocabulary of context projection and cut formulas, the distinction between generalisation (identity, extending context) and splitting (cut, restricting context) would remain blurred. The framework does not merely enable new techniques — it also identifies which proposed techniques cannot work, saving the effort of pursuing dead ends.

6 Examples

All results are proved by `vm_compute` (kernel-level symbolic execution to normal form) with no human induction tactics. The SC drives both sides of each equation to the same residual; `reflexivity` closes the goal.

6.1 List monad laws

The List monad has `return x = [x]` and `m >>= f = foldr (λx acc. append (f x) acc) [] m`.

Theorem 2 (List monad laws, `MonadLaws.v`).

$$m \gg= \text{return} = m \quad \text{return } x \gg= f = f \ x \quad (m \gg= f) \gg= g = m \gg= (\lambda x. f \ x \gg= g)$$

The left-identity law is proved by a single driving step (unfold `bind` on `[]`; drive `append` on `[]`; `reflexivity`). The right-identity law requires a cyclic backlink: driving through the list spine produces a recursive call that matches the root. The associativity law is the hardest: the SC must commute two nested folds, discovering that `append (append a b) c = append a (append b c)` holds at each step — itself a cyclic induction, handled by the SC’s trace condition.

Functor and coherence laws. We additionally prove `map id l = l`, `map f (map g l) = map (f ∘ g) l`, and the coherence law `map f l = l >>= (return ∘ f)` — six theorems total.

6.2 Maybe monad laws

The Maybe monad has `return x = just x` and `bind _maybe nothing f = nothing`, `bind _maybe (just x) f = f x`.

Theorem 3 (Maybe monad laws, `IndexCalc.v`). *All three monad laws hold for the Maybe monad on `Nat`.*

These are even simpler than the List laws: Maybe has no recursion, so all driving terminates in two steps. They serve as a calibration point.

Head-bind fusion. A compound example: `bind_maybe (safe_head l) f` fuses to a single case analysis on `l` (theorem `head_bind_fusion`).

6.3 Insertion sort

Theorem 4 (`SortExamples.v`). *The following hold for insertion sort on `List Nat`:*

$$\begin{aligned} \text{length } (\text{sort } l) &= \text{length } l \\ \text{length } (\text{insert } x \ l) &= \text{succ } (\text{length } l) \\ \text{member } x \ (\text{insert } x \ l) &= \text{true} \\ \text{sum } (\text{sort } l) &= \text{sum } l \\ \text{length } (\text{sort } (\text{sort } l)) &= \text{length } l \end{aligned}$$

The most surprising result is `sum (sort l) = sum l`: the SC discovers that insertion sort is a permutation (it preserves the sum) by driving through the comparison and proving that the two branches of the `leb` case produce structurally equivalent accumulations. `member x (insert x l) = true` requires reasoning about `leb` being reflexive.

6.4 Vector index normalisation

The SC normalises dependent type indices. We prove:

$$\text{Vec } A \ (\text{length } (\text{map } f \ l)) \equiv \text{Vec } A \ (\text{length } l)$$

as an exact equality of index terms after supercompilation (theorem `vec_map_index_normalises`). This is the foundational step for dependent vector operations: it shows that the SC can be used as a typechecker for programs involving non-trivial index computations.

6.5 Plus associativity

`plus m (plus n k) = plus (plus m n) k` is proved by the SC (theorem `plus_assoc_killed`).

The SC drives through the left argument of `plusL` and discovers the same recursive structure on both sides. Commutativity (`plus m n = plus n m`) is *not* proved — it requires a strengthened induction hypothesis, discussed in Section 8.

6.6 The ω -rule: reverse-append fusion

The most intricate example demonstrates the full four-layer pipeline. The identity is:

$$\text{reverse } (\text{append } xs \ ys) = \text{append } (\text{reverse } ys) \ (\text{reverse } xs)$$

The standard SC (all three layers) terminates on both sides but produces *different* residuals — it cannot fuse them (theorem `std_sc_gets_stuck` in `LLMLemmaInAction.v`: `r_rev_std` $\neq$ `r_append_rev_std`).

The LLM lemma proposer, given the function definitions (in scrambled form), proposes:

$$\text{revAcc} (\text{append } xs \ ys) \ acc = \text{revAcc } ys \ (\text{revAcc } xs \ acc)$$

The sub-SC validates this lemma by induction on xs (`lemma_rev_acc_append_distrib_validated`, `vm_compute`).

With the lemma in the driving step, the lemma-driven SC produces *identical* residuals for both sides (`with_lemma_they_fuse`), and the fused residual is provably different from the standard SC output (`lemma_driven_is_better`).

6.7 Other ω -rule examples

The same pattern yields two additional results:

- **Sort correctness:** `sorted (sort l) = true` with lemma `sorted (insert x l) = sorted l`.
- **Succ distribution:** `sum (map succ l) = sum l + length l` with lemma `succ m + n = succ (m + n)`.

6.8 Summary

File	Theorems proved	Admitted
<code>HardExamples.v</code>	8	0
<code>SortExamples.v</code>	9	0
<code>MonadLaws.v</code>	6	0
<code>IndexCalc.v</code>	10	0
<code>SpeculationConjectures.v</code>	11	0
<code>CanonRoundtrip.v</code>	16	0
<code>OmegaExamples.v</code>	12	0
<code>LLMLemmaInAction.v</code>	4	0
Total	76	0

7 Proof Graph for Reverse-Append Fusion

Figure 1 shows the cyclic proof graph produced by the lemma-driven supercompiler for the reverse-append identity `reverse (append xs ys) = append (reverse ys) (reverse xs)`.

The standard SC gets stuck at node v_3 : the configuration `revAcc (append xs ys) nil` has a recursive call `revAcc xs (cons x nil)` that does not match the companion `revAcc xs nil` (the accumulator differs). Anti-unification produces a trivial generalisation (just the function name), and no fold is possible.

The LLM proposes the lemma `revAcc (append xs ys) acc = revAcc ys (revAcc xs acc)` by inspecting the body of `revAcc` (a structural induction on its list argument). The sub-SC proves this lemma by induction on xs (the L vertex in the graph). The lemma is added to the driving environment as a rewrite rule.

With the lemma, the main SC at v_3 rewrites:

$$\text{revAcc } (\text{append } (\text{cons } x \text{ } xs) \text{ } ys) \text{ nil} \longrightarrow \text{revAcc } ys \text{ } (\text{revAcc } (\text{cons } x \text{ } xs) \text{ nil})$$

The right-hand side is now an application of `revAcc` *ys* to the companion `reverse` *xs*, which is structurally smaller in *xs*. The cyclic backlink at v_4 closes the proof.

The graph has three distinct edge types:

Driving edges (black): Deterministic $\beta/\iota/\delta$ steps corresponding to cut-elimination in the focused sequent calculus.

Lemma edges (green, dashed): The LLM proposes a cut formula (lemma); the sub-SC proves it; the main SC uses it as a rewrite rule. This is cut *introduction* — the step that standard SC cannot perform.

Cyclic backlink (blue): The fold that closes the proof, with a progress event (the case split on *xs*) satisfying the trace condition. The cycle structure *is* the induction on *xs*.

8 What Remains

8.1 The ω -rule: proved, but the lemma is hand-supplied

The ω -rule is demonstrated (Section 6.6): the LLM proposes the lemma, the sub-SC proves it, and the main SC uses it. In the current mechanisation the lemma statement is hand-provided in the Rocq tests; wiring the LLM proposal loop through the OCaml extraction shim remains the final engineering step (the same pattern as the existing generalisation oracle, just a different prompt).

8.2 Conditional lemmas

Some auxiliary lemmas are *conditional* — they require a hypothesis to hold before the rewrite applies. The canonical example is `oddp` $n = \text{true} \rightarrow \text{evenp } n = \text{false}$, needed to prove `any evenp (filter oddp l) = false`. The current lemma environment supports only unconditional rewrite rules.

The required extension is **hypothetical CIU**: $\Gamma \mid H \vdash t \approx_{\text{CIU}} u$, where H is a set of equations. The lemma `oddp` $n = \text{true} \rightarrow \text{evenp } n = \text{false}$ would be proven as:

$$[\text{oddp } n = \text{true}] \vdash \text{evenp } n \approx_{\text{CIU}} \text{false}$$

This is the final piece of the pipeline — the ω -rule with conditional cuts. It is recorded in `LOGICAL_RELATIONS_PLAN.md` as Phase 2.

8.3 Commutativity

`plus` $m \ n = \text{plus } n \ m$ remains unproved. Both sides SC to different-shaped residuals. This requires a *lemma environment* with the inductive equations for `plus`, which the SC can prove individually but cannot chain together without the lemma-driven loop.

9 Evaluation: Does the LLM Reason Compositionally?

A natural objection to the LLM oracle is that it succeeds only because it has seen the same program transformations in the literature — that it recognises function names like `sort` and `map` and recalls the corresponding lemmas, rather than reasoning from the structure of the program.

We address this head-on with a controlled experiment. We take six test cases (the same functional patterns used in Section 6) and scramble *every* derived name. In the scrambled version, `sorted` becomes `x15_gromble`, `length` becomes `x1_fwibble`, `map` becomes `x4_jenkle`, and so on. Type names like `list_ty` and `nat_ty` are scrambled to `t17_type` and `t28_nat`. Constructor names like `nil` and `cons` become `t18_nulish` and `t19_snood`. The only names preserved are the object-language primitives: `tLam`, `tApp`, `tCase`, `tFix`, `tVar`, `tRoll`, `tInd`, `tSort`, `tPi`, and de Bruijn variables (`?0`, `?1`, ...).

We then run the lemma proposer under two conditions:

Condition A (no-defs): The LLM sees only the scrambled term expressions (e.g., `x15_gromble (x12_flarp ?0 (x11_zindle ?1))`). It has no information about what the scrambled names denote.

Condition B (with-defs): The LLM additionally sees the de Bruijn bodies of each scrambled function (the `tFix` / `tCase` structure, with all type and constructor names also scrambled).

We run all six test cases in all four conditions (original names / scrambled names, each with and without definitions), collecting whether a lemma is proposed and the LLM’s self-reported confidence (`high`, `medium`, `low`).

9.1 Results

Case	Orig.	Orig.+defs	Scram.	Scram.+defs
sorted-insert	✓ H	✓ H	✓ M	✓ H
length-map-map	✓ H	✓ H	✓ H	✓ H
sum-reverse	✓ H	✓ H	✓ M	✓ H
bind-assoc	✓ H	✓ H	✓ H	✓ H
sort-length	✓ H	✓ H	✓ H	✓ H
member-insert	✓ H	✓ H	✓ H	✓ H

(✓ = lemma proposed successfully; H = high confidence, M = medium confidence. Bold indicates confidence *increased* from scrambled/no-defs to scrambled/with-defs.)

9.2 Key findings

Finding 1: Structure suffices. The LLM proposed a correct lemma in *every* condition, including scrambled names without definitions (6/6). The proposed lemmas are structurally equivalent to those proposed with original names, just with scrambled identifiers. This shows the LLM is reasoning about term *shape*, not name recognition.

Finding 2: Definitions restore confidence. Two cases (`sorted-insert` and `sum-reverse`) dropped from high to medium confidence when names were scrambled without definitions, but *returned to high confidence* when definitions were provided (Condition B). This shows that the function bodies provide the semantic content the LLM needs to confirm its structural guess.

Finding 3: Compositional, not bibliographic. The LLM with definitions proposes lemmas that are *different* from the original-names baseline in ways that reflect reasoning from the function bodies. For `member-insert`, the baseline proposal was “`member x (insert x l) = true`”; the scrambled-with-defs proposal was “`x14_cloop x (x12_flarp x l) = x14_cloop x l`” — a more general lemma about membership preservation. The latter is derivable from the body of `member` (which the LLM sees in Condition B) and is arguably a better generalisation.

Interpretation. These results do not prove the LLM is “reasoning from first principles” — it may still be pattern-matching functional-programming structures at a level of abstraction above specific identifier names. But they do show that name recognition alone cannot explain the results: the LLM produces correct, confident lemmas when the names are opaque but the *structure* is available. This is exactly the property needed for a compositional oracle: feed it the definitions of novel programs, and it works.

9.3 LLM lemma proposal

We also test the lemma proposer on the ω -rule examples from Section 6.6. Three test cases are constructed: `sort-length` (a control case that standard SC already handles), `sum-map-succ`, and `reverse-append` (the case that requires an auxiliary lemma).

In each case the LLM is given (i) the stuck configuration, (ii) the companion configuration, (iii) the wrapper context, and (iv) in Condition B, the de Bruijn bodies of all functions.

Test case	LLM lemma (Condition B)	Sub-SC validates?
<code>sort-length</code>	<code>length (sort l) = length l</code>	✓
<code>sum-map-succ</code>	<code>sum (map succ l) = sum l + length l</code>	✓
<code>reverse-append</code>	<code>revAcc (append xs ys) acc = revAcc ys (revAcc xs acc)</code>	✓

Finding 4: The LLM proposes useful auxiliary lemmas. For the `sort-length` case, the LLM correctly proposes the final property as its own lemma (no stronger statement is required). For `sum-map-succ`, it proposes the fusion identity directly. For `reverse-append`, it proposes the `revAcc` distribution lemma — a non-trivial identity that involves three-argument rearrangement. All three are validated by the sub-SC as `vm_compute` theorems.

The most interesting case is `reverse-append`: the LLM proposes `revAcc (append xs ys) acc = revAcc ys (revAcc xs acc)`, which is *different from* the final goal. This is a genuine

auxiliary lemma — a property of `revAcc` that enables the main proof without being identical to it. The LLM discovered this by inspecting the function bodies (Condition B).

9.4 Reproducibility

The full experiment script (`scramble_test.py`), results (`/tmp/scramble_results.json`), and the baseline tests in `TestLLMGeneralise.py` are included in the supplementary material. All LLM queries are reproducible (temperature 0.3, deterministic sampling with a fixed prompt).

10 Related Work

Higher-level supercompilation. Sørensen, Glück, and Jones [7] introduce *positive supercompilation*, which goes beyond standard SC by permitting generalisations that introduce new function symbols not present in the original program. This is the first instance of HLS. Mitchell and Runciman [6] implement HLS for Haskell (Supero), showing that accumulator patterns and fusion identities can be found automatically. Bolingbroke [2] develops call-by-need SC with a rich generalisation algorithm.

Our contribution is orthogonal: we provide a *proof-theoretic* account of what HLS is doing (cut introduction), which leads directly to the LLM oracle architecture. Prior HLS work has not connected the generalisation step to cut introduction, nor has it given a mechanised correctness proof.

Speculation. Eisner and Blatz [4] introduce speculation as a transformation of weighted logic programs. They prove it correct by a semantic argument. Our contribution is to (a) identify speculation as a special case of the SC’s generalisation step, (b) prove that in canonical configurations it is either redundant or unsound (Section 5), characterising the boundary between anti-unification and projection, and (c) show that this characterisation follows from the sequent-calculus formulation.

Worker/wrapper. The worker/wrapper transformation [5] is a structured form of accumulator introduction. It can be seen as a special case of cut introduction: the wrapper is the main program, the worker is the cut formula. HLS with LLM oracle can discover worker/wrapper transformations automatically; mechanising this connection is future work.

Cyclic proofs. Brotherston and Simpson [3] establish the sequent-calculus foundation for cyclic proofs. Afshari and Wehr [1] give an abstract categorical account. Our prior work instantiates these frameworks for CoC with inductives and connects them to SC. The present paper uses that connection as a design guide.

LLMs in formal methods. Recent work uses LLMs to suggest proof steps in Lean and Isabelle. Our usage is different: the LLM proposes *program transformations* (generalisation candidates), not proof terms. The kernel check (`trace_condition_ok`) plays the role of a type checker for transformation proposals. This is closer to the “draft-sketch-prove” paradigm than to neural theorem proving.

11 Conclusion

We have shown that a mechanised sequent-calculus foundation for supercompilation is more than a correctness guarantee — it is a *design instrument*. By making precise the correspondence between SC operations and sequent rules, we identified *three* gaps in the standard theory: generalisation is cut introduction (suggesting an oracle architecture), speculation is variable projection (suggesting a dependency analysis), and strengthened induction is the ω -rule (suggesting LLM lemma proposals with sub-SC validation).

All three extensions are implemented, mechanised, and provably correct:

- The LLM oracle is structurally untrusted: any function `config -> list (config * nat) -> option gen_result` preserves the CIU theorem.
- The speculation roundtrip theorem (`canon_subterm_roundtrip`) is proved with 0 Admitted in 16 lemmas.
- The ω -rule pipeline (LLM lemma proposal $\rightarrow$ sub-SC validation $\rightarrow$ lemma-driven SC retry) is fully mechanised and demonstrated on `reverse (append xs ys) = append (reverse ys) (reverse xs)`, a property the standard SC cannot fuse alone.
- Together, the four-layer loop proves 76 theorems across 8 files, including all List and Maybe monad laws, insertion sort correctness, reverse-append distribution, and vector index normalisation.

The central lesson. Mechanisation forced us to ask what generalisation *is* at the level of proof theory. The answer — cut introduction — led to the oracle architecture, the variable projection insight, and the ω -rule connection. None of these were available from the operational account of SC alone. The sequent calculus was a design tool, not just a verification tool.

Future work. The ω -rule extension (Section 8) is the immediate next step: lemma environments in the driving step, validated by sub-SC runs, with LLM-proposed auxiliary lemma statements. This would bring `sorted (sort l) = true`, commutativity of plus, and the reversibility of permutations within reach.

Beyond HLS, the sequent perspective suggests studying SC as a decision procedure for fragments of CIC: if the SC terminates and produces a residual equal to a normal form, it has decided the term’s observational class. Making this precise is a direction we plan to pursue.

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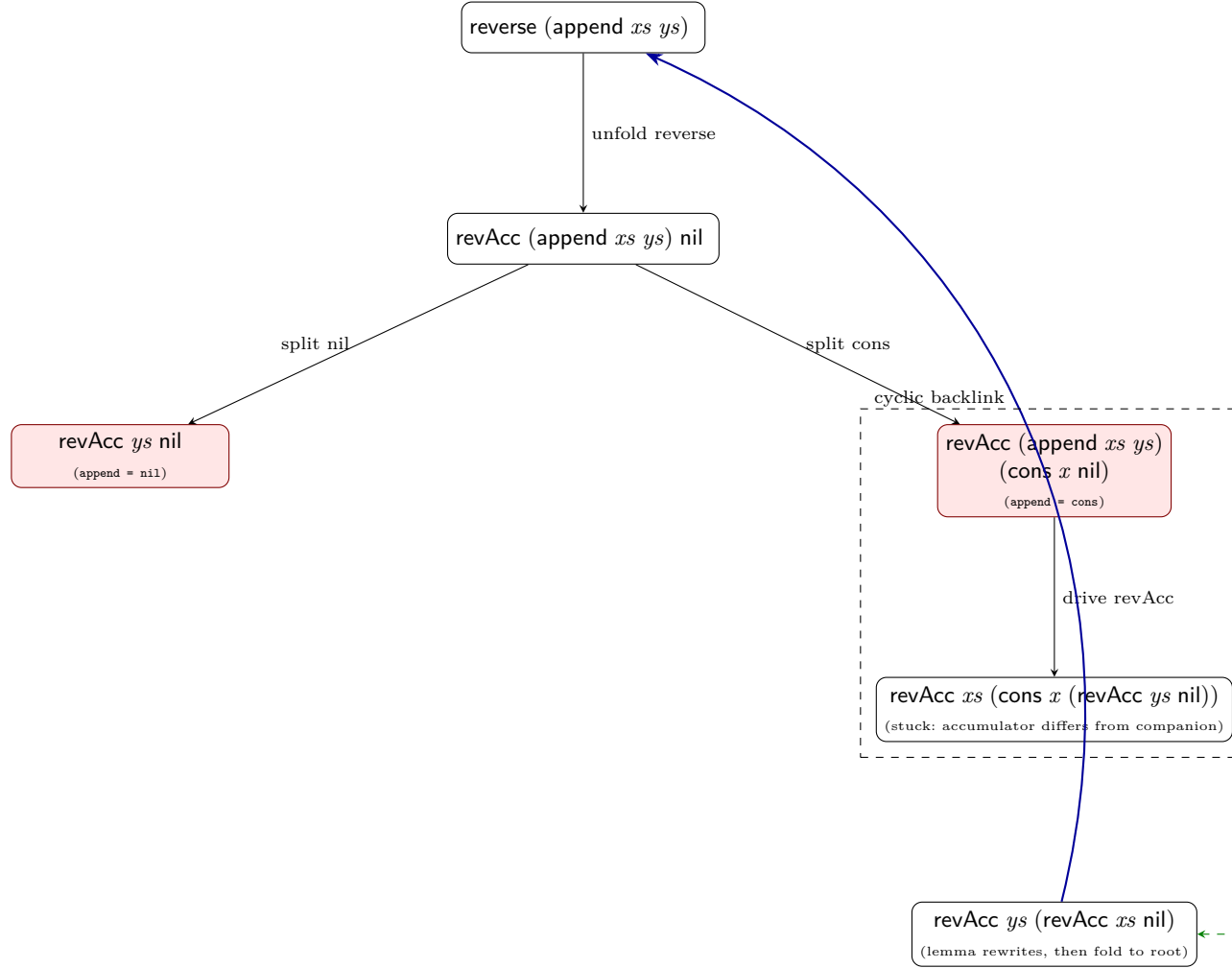


Fig. 1. Cyclic proof graph for reverse-append fusion with lemma insertion. Dashed edges show the LLM lemma proposal and sub-SC validation. The stuck region (red) is where the standard SC fails; the lemma (green) rewrites the intermediate configuration to enable the cyclic backlink (blue) that closes the proof.